Óscar Fraile

Full Stack Engineer

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Profile

I'm a software engineer who enjoys working on the backend, figuring out how the pieces fit together, and turning ideas into working products. I also care about thoughtful design and making those products intuitive to use. Lately, I've been building tools with AI.

Professional experience

Full Stack Engineer | Educaria

2019 - Present · Madrid, Spain

Building **Alexia** - the leading EdTech platform for schools and universities in Spain and Latin America, with **2M+** active users across web, Android, and iOS.

- Contributing to the transition from a .NET Framework 4.8 monolith to distributed **microservices** with **.NET Core** and Entity Framework Core, applying Hexagonal Architecture, **Clean Architecture**, SOLID, DDD and CQRS.
- Full stack development with **React** and **TypeScript**, authentication with **Identity Server** and caching with **Redis**.
- **CI/CD** pipelines in Azure DevOps, **Docker** containers and deployments to **Kubernetes** managed with Rancher.
- Unit tests with **xUnit**, integration tests, and API and load-testing scenarios with **k6**.
- AI-assisted development with **Claude Code**, **MCP** and shared skills, using **Spec-Driven Development** to guide implementation and review with clear requirements and acceptance criteria.
- Development of a **RAG** chatbot with **Python and FastAPI** for teachers and students, supported by knowledge of chunking, embeddings, metadata, reranking and grounding, as well as agent orchestration with **LangChain and LangGraph**, query rewriting and context management to balance response quality, latency and cost.

Solutions Assistant | NTT Data

2017 - 2019 · Madrid, Spain

Software developer on a large-scale project for **Repsol**, powering point-of-sale across gas stations in Spain and Peru.

- Built mission-critical point-of-sale, handheld, and back-office systems.
- Joined the early phase of the new back-office front-end, built with **Angular**.
- Handled development, unit testing, and documentation within agile teams.

Technical skills

Frontend: React, TypeScript, Angular, Ionic, JavaScript, Tailwind CSS

Backend: C#, .NET, Entity Framework Core, Python, FastAPI, SQLAlchemy, Identity Server, Node.js, OData, GraphQL

Architecture & DevOps: Microservices, Hexagonal Architecture, Clean Architecture, SOLID, Docker, Kubernetes, Rancher, CI/CD, Azure DevOps, GitHub Actions, Azure, Google Cloud, Vercel

Databases: SQL Server, PostgreSQL, Redis, Firestore, MongoDB

AI & Automation: Claude Code, MCP, Spec-Driven Development, RAG, LangChain, LangGraph, GitHub Copilot, Prompt Engineering, AI Agents

Testing & Quality: xUnit, Moq, k6, BDD / Gherkin, SonarQube

Selected projects

GolA

Speaker diarization platform - from the model to the API

Async, job-based Batch API in **FastAPI** on **Cloud Run**: create job → upload audio → run → poll/webhook → results.

On-demand GPU orchestration: spins up ephemeral **Compute Engine** VMs (T4→L4 zone cascade) that run the model in **Docker** and self-delete when done.

2026 World Cup Predictor

Match predictions for the 2026 World Cup, with an honest track record

Two-phase **Poisson GLM**: a pre-trained prior from international matches, blended with in-tournament learning via shrinkage as more games are played.

Dixon-Coles correction for low-scoring matches and a full scoreline matrix to derive 1X2 probabilities and favorites.

Education

Bachelor's Degree in Computer Engineering | 2026

UDIMA - Universidad a Distancia de Madrid

Higher Diploma in Multiplatform Application Development | 2017

Centro Europeo de Estudios Profesionales

Certifications

Microsoft Certified: Azure Developer Associate · Microsoft · 2026

Claude Code in Action · Anthropic · 2026

Machine Learning with Python · IBM · 2026

Mathematics for Understanding the World: Models, Data, and Artificial Intelligence · UDIMA - Universidad a Distancia de Madrid · 2026

What The Hack: Azure OpenAI Fundamentals · Código Facilito · 2024

Cambridge English: Advanced (CAE) - CEFR C1 · University of Cambridge · 2018

Languages: Native Spanish · English C1 (Cambridge CAE)